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Thumb-actuated locking holster

Abstract

A holster for receiving and withdrawing a handgun has a proximal wall portion with a handgun retention mechanism thereon. The mechanism having a first lever with a tab portion for actuation and release of a trigger guard, the lever having a snap fit into a lever region without a separate pin. A belt engaging member covers the lever region capturing the first lever therein. If a second retention mechanism is used, another lever with a tab portion is positioned so the tabs are stacked, depressing one also depresses the tab behind it, such that both mechanisms are released. Overmolded polymer layers in the holster body provides a softer interior handgun engagement surfaces. The holster body is split forwardly with a threaded fastener for clamping onto the handgun. No holster body is beneath or rearward of the tab portion, configured as a thumb button, allowing smooth transition to gripping the handgun.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/245,688 filed Apr. 30, 2021, which is a continuation of U.S. Non-Provisional patent application Ser. No. 16/748,151 filed Jan. 21, 2020 now U.S. Pat. No. 10,996,024, which is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 16/364,141 filed Mar. 25, 2019 now U.S. Pat. No. 10,619,974, which is a continuation of U.S. Design patent application Ser. No. 29/647,062 filed May 9, 2018, now U.S. Patent No. D860641 and claims the benefit of U.S. Provisional Application Nos.: 62/647,265, filed Mar. 23, 2018; 62/712,717 filed Jul. 31, 2018 and 62/794,594 filed Jan. 19, 2019, the entire contents of which are incorporated by reference herein.

BACKGROUND OF THE DISCLOSURE

- (1) In many countries (e.g., Germany, Poland and the United States) law enforcement officers often carry a handgun in an openly visible manner while on duty. The handgun is typically carried in a holster designed to protect the handgun and hold it securely. The holster may include a belt loop that allows it to be hung from a wide belt, for example, a police officer's service belt. The holster may also include a locking mechanism to prevent an assailant from drawing the holster user's weapon and prevent the handgun from inadvertently falling out of the holster, for example, when the holster user is running, climbing over a fence, etc. It is important that the retention mechanisms are secure in such challenging situation.
- (2) In addition to having the handgun be secure in the holster in challenging situations, ease of withdrawal when urgently needed by the user is an important feature. That is, easy actuation of release mechanisms in stressful situations.
- (3) A holster that securely retains a handgun and that provides improvement in ease of use when the handgun is urgently needed would be welcomed by the industry.
- (4) To have holsters be commercially viable, they need to be priced to meet expectations of the consuming populace. So any improvement of assembly and cost efficiencies by minimizing the

number of parts and minimizing the number of assembly steps would also be welcomed by the industry.

SUMMARY

(5) A holster for receiving and holding a handgun having a trigger guard comprises a polymer holster body and an elongate polymer pivoting lever that together define a handgun retention mechanism that is thumb actuated and moves a trigger guard block portion of the lever between retention and release positions. A thumb button depressible toward the handgun is optimally positioned on the user side of the holster, is framed by the holster body and is shielded by thumb guides which may be projecting portions of the body or by components both above and below (with respect to the handgun orientation) the thumb button. In the same motion of grasping the handgun the thumb button is depressed releasing the handgun. A feature and advantage of this arrangement is that unintended access to the handgun release, specifically the thumb button, particularly by others, is limited, while allowing intuitive and easy user access to the thumb button by user's thumb when grasping the handgun.

(6) Although thumb buttons that release trigger guard block portions are known, such buttons are not well protected from inadvertent actuation, or actuation by others and could use improvement in intuitively accessing such buttons by the user.

(7) In embodiments, the holster may be a level-2 configuration, with one active restraint, for example the trigger guard, or a level-3 configuration, with two active restraints. Both the level 2 configuration and the level 3 configuration utilize the optimally positioned thumb button and may include the elongate polymer pivoting lever that selectively engages the trigger guard of the handgun. The level-3 holster configuration also includes a hood or shroud, such as a U-shaped shroud, that is pivotally supported by the holster body so that the U-shaped shroud selectively pivots between a handgun retaining position in which the U-shaped shroud extends across a rearward opening of the cavity and a release position in which the U-shaped shroud is displaced from the rearward opening so that a withdrawal path of the handgun is unobstructed.

(8) In embodiments, a thumb button of the retaining mechanism is positioned to allow for an intuitive and ergonomic release of the handgun from the holster by positioning the thumb button on the user's side of the holster exactly forward (with respect to the handgun) of the upper handgrip or backstrap recess; the handgrip or backstrap recess is located below the handgun slide of holstered handgun. The thumb button is sized and positioned to receive the inside surface of the thumb at the distal knuckle. The thumb button is actuated by depressing it toward the handgun and is spaced about one half inch or less from the handgun body. When a user lowers his hand onto the handle of the holstered handgun, with the webbing between the user's thumb and forefinger engaging the backstrap recess, as the user grasps the handgun grip, he will wrap his middle, ring and pinky fingers around the grip of the handgun, will move the thumb naturally to an actuation position with respect to the thumb button, and will allow the forefinger to be received by an elongate finger recess extending forwardly (with respect to the handgun) on the holster. Squeezing the grip for a full grasping of the handgun can readily depress the thumb button and release the handgun. This arrangement advantageously allows the user to actuate the thumb button and release the handgun with much less or no concerted effort of depressing a release button. While the forearm is extending in a downward direction, grasping of the holstered handgun is with less tension in the muscles and tendons of the thumb and forearm. The thumb button position as described provides for natural and comfortable movement of the thumb, the same movement as grasping the handgun. It is noted that the mechanics of the muscles and tendons of the thumb and forearm when downward readily accomplish this grip and handgun release, the mechanics change when the forearm is extending in a horizontal direction away from the torso of the handgun user. The master grip is facilitated in embodiments, both level two and level three, where the thumb button has no holster body portions or other holster components directly rearward of the rearward margin of the thumb button, and no holster body portions below the thumb button. Such embodiments provide

the feature and advantage that as the handgun is withdrawn, the user's thumb slides off of the thumb button into immediate contact with the handgun. That is, there is no engagement with the exterior surface of the holster body by the user's thumb either during actuation of the release of the retention mechanism or as the handgun is withdrawn. In that the thumb button is positioned in close proximity to the surface of the handgun, the thumb movement inward as the thumb transitions from engagement with the button to engagement with the body of the handgun is minimal.

(9) In embodiments, a holster for receiving a handgun having a trigger guard comprises a holster body having a plurality of holster wall portions defining a rearward opening cavity with a handgun receiving and withdrawal axis. The holster has a first handgun retention mechanism, a second handgun retention mechanism, and a thumb receiving tab that actuates both the first handgun retention mechanism and the second handgun retention mechanism. In embodiments, each handgun retention mechanism selectively prevents the handgun from being withdrawn from the cavity defined by the wall portions of the holster body.

(10) In embodiments, the first handgun retention mechanism may comprise a lever pivotally supported by the holster body. In embodiments, the lever has an elongate central portion, a depressible portion, and a blocking portion. The depressible portion of the lever comprises a first arm extending away from the elongate central portion in a first direction so that the elongate central portion and the first arm cooperate to form a first L-shape. The blocking portion of the lever comprises a second arm extending away from the elongate central portion in a second direction opposite the first direction so that the elongate central portion and the second arm cooperate to form a second L-shape. The entire lever having a stretched Z shape. In embodiments, the lever is pivotally supported by the holster body so that the lever pivots about a lever axis that extends in upward and downward directions with respect to the worn holster, and forward and rearward with the handgun orientation. In embodiments, the lever pivots between a handgun trigger guard capture position and a handgun trigger guard release position. The blocking portion of the lever extends into a trigger guard portion of the holster body while the lever is disposed in the handgun trigger guard capture position.

(11) In embodiments, the second handgun retention mechanism may comprise a hood or shroud, for example a U-shaped hood or shroud that is pivotally supported by the holster body so that the U-shaped shroud selectively pivots between a handgun retaining position in which the U-shaped shroud extends across a rearward opening of the cavity and a release position in which the U-shaped shroud extends forward of the rearward opening so that a withdrawal path of the handgun is unobstructed.

(12) In embodiments, the holster includes a U-shaped shroud that is part of a hood assembly including a bracket that captures the U-shaped shroud and defines a slot. In embodiments, the holster includes a leaf spring that is received in the bracket slot when the U-shaped shroud is in the handgun retaining position. A spring biases the U-shaped shroud to pivot toward a handgun withdrawal position. In embodiments, depression of the thumb button causes the lever to rotate through a first range of rotary motion and causes deflection of the leaf spring so that a distal portion of the leaf spring is outside of the slot. In embodiments, further depression of the thumb button causing rotation of the lever through a second range of rotary motion causes the thumb button to engage the depressible portion of the lever causing at least part of the blocking portion of the lever to be withdrawn from the trigger guard receiving portion of the holster body releasing the trigger guard and allowing removal of the handgun.

(13) In embodiments, two handgun retention mechanisms utilize pivoting components rather than sliding components, this is believed to generally reduce the size, complexity, and reliability of the mechanisms and components. Additionally the two actuatable handgun retention mechanisms, and particularly the pivoting components, are mounted on the exterior of the holster body, thereby simplifying assembly, operation, maintenance, and cleaning, if needed.

(14) In embodiments, the holster body is formed from two polymers, one overmolded onto the other, providing an interior layer and an exterior layer. In embodiments, the interior layer being softer than the exterior layer minimizing any wear, damage or markings on the exterior surface of the handgun. In embodiments, the interior layer and the exterior layer are combined using an injection molding “overmolding” process where one layer is injection molded onto the previously molded layer. In embodiments, the interior layer comprises a thermoplastic elastomer (TPE) and the exterior layer a thermoplastic material. In embodiments, the interior layer comprises a polymer such as Hytrel and the exterior layer comprises a polyamide material (e.g., nylon).

(15) In embodiments, the holster body is formed to provide a handgun fitting function that can be adjusted by rotating a tensioning screw. The shape of the forward portion of the handgun can be conceptualized as a rectangle and the shape of the cavity defined by the holster body can be configured to make point contact with the four corners of the rectangular handgun portion. In embodiments, the shape of the cavity defined by the holster body can be conceptualized as two opposing V-shapes, an upper V-shape and a lower V-shape. The two legs of the upper V-shape and the two legs of the lower V-shape may each contact one corner of the rectangle. In embodiments, the lower V-shape defines a split near the lower end of the V-shape. A tensioning screw is positioned to selectively decrease the angle between the two legs of the lower V-shape. As the angle between the two legs of the lower V-shape decreases, the two legs of the lower V-shape apply upwardly directed component forces to the corresponding corners of the rectangular handgun portion. The forces applied to the lower corners of the rectangular handgun portion urge the upper corners of the rectangle against the two legs of the upper V-shape. In embodiments, the softer inner layer provides a higher level of friction to the insertion and withdrawal of the handgun and thus the adjustment of the tensioning screw is more sensitive and effective in controlling the friction associated with a handgun withdrawal compared to the same configuration with a harder plastic interior surface.

(16) A feature and advantage of embodiments is a retention mechanism has two distinct separated pivoting connections forming a hinge portion of the handgun retention mechanism. The hinge portion of the retention mechanism has a hinge length extending between the outer end portions of the two pivot portions. A ratio of the hinge length to the overall length of the elongate lever is greater than 0.8. This arrangement provides stability of attachment, robustness and ease of assembly. This arrangement also places less stress on the holster body. In embodiments, the pivot portions of the lever and pivot of the body are coupled using a simple assembly process with no separate hinge pin.

(17) A rearward pivoting connection and a forward pivoting connection of the retention mechanism may each be formed from a pair of cooperating connector pivot portions. In embodiments, a forward pivot portion is a pin portion and the other forward pivot portion is a C-shaped pin receiving portion. One of the rearward pivot portions may be a pin portion and the other of the rearward pivot portions may be a C-shaped pin receiving portion. One of the rearward pivot portions may be integrally formed with the holster body and the other of the rearward pivot portions may be integrally formed with the lever. One of the forward pivot portions may be integrally formed with the holster body and the other of the forward pivot portions may be integrally formed with the lever.

(18) A feature and advantage of embodiments is a pivoting arrangement having a pin portion that snaps into a corresponding C-shaped portion. This arrangement provides manufacturing advantages including fewer parts, easier assembly, the possibility of performing a repair by replacing the lever, and the possibility of replacing the lever with one or more levers having alternate configurations to suit the preferences of different users.

(19) A feature and advantage of embodiments is a pivoting handgun release arrangement in a holster made almost entirely of polymeric material, except for springs and fasteners. This arrangement provides ease of assembly and cost efficiencies by minimizing the number of parts

and minimizing the number of assembly steps. In embodiments, the pivoting arrangement includes a polymer pin portion that is received in a corresponding C-shaped portion.

(20) A feature and advantage of embodiments is a retention mechanism in a holster that is thumb actuated to selectively release the handgun when the user wishes to draw his or her weapon. The release actuation mechanism includes a pivoting lever. The blocking portion of the lever pivots from the blocking position to the release position when the user's thumb applies a pivoting force to the thumb receiving portion of the lever.

(21) A feature and advantage of embodiments is that thumb actuating release actuation mechanism is biased, such as by a spring, to a preactuation position and is automatically reset after withdrawal of the handgun. The handgun can be reholstered without manual reset of the retention mechanism or the release actuation mechanism.

(22) A feature and advantage of embodiments of a holster and attached belt loop is that the release button of the holster is positioned between the handgun and dy and a belt loop and the holster user and nested within a three-sided frame of the holster body. The thumb release button in this position is not readily accessible or visible to potential attackers.

(23) A feature and advantage of embodiments is a thumb actuated holster with a depressible thumb button, the holster mounted to a jacket slot belt loop, the jacket slot having vertical ribs that inhibit lateral access to the depressible thumb button.

(24) A feature and advantage of embodiments is a thumb-actuated lever that is captured between the holster and a mounting plate defining one or more slots for receiving belts, straps, releasable holster attachment systems, and the like. In embodiments, the holster body and the mounting plate define a funneling portion that may be guide the user's thumb toward the thumb-actuated lever.

(25) A feature and advantage of embodiments is a holster having two handgun retention mechanisms that are both actuated by a single thumb-actuated button. In embodiments, one of the handgun retaining mechanisms comprises a U-shaped shroud that is selectively positionable to extend across an upper opening of the holster body. In embodiments, the other of the handgun retaining mechanisms comprises a lever having a trigger guard block portion. In embodiments, the U-shaped shroud and the lever are actuated sequentially by a single thumb-actuated button. In embodiments, the U-shaped shroud is released before the lever is rotated. In embodiments, releasing the U-shaped shroud before the lever is rotated assures that the U-shaped shroud is out of the way before the user attempts to withdraw the handgun from the holster.

(26) A feature and advantage of embodiments is a holster having a U-shaped shroud that may remain open when the handgun is reholstered, allowing the holster to be used in a mode having a lesser level of retention than with the U-shaped shroud in a handgun obstructing position. In embodiments, the U-shaped shroud may be reset by rotating the U-shaped shroud to a position in which the U-shaped shroud obstruct the withdraw path of the handgun.

(27) A feature and advantage of embodiments is a holster comprising one or more finger guides that facilitate consistent and intuitive gripping of the handgun before, during and after drawing the handgun from the holster. In embodiments, the finger guides guide the user's hand to consistently hold the handgun in the master grip upon drawing the handgun from the holster. In embodiments, the hand assumes the master grip position while the handgun is still holstered in the holster. Thumb guides direct the user's thumb to a thumb button and a datum element is located so that the index finger of the hand is received in the groove while the grip portion of the handgun is being gripped in the palm of the hand and the index finger is extending downwardly away from the palm of the hand. In embodiments, the handgun is held in the master grip after the handgun is unholstered. The index finger may transition from holster engagement to handgun engagement as the handgun is withdrawn from the holster cavity. The thumb easily transitions from the thumb button to gripping the handgun body. In embodiments, the holster comprises a novel combination of elements that facilitate reliable and intuitive unholstering of the handgun.

(28) A feature and advantage of embodiments is that the groove defined by a datum element on the

outside panel of the body provides a tactile indication of where the index finger should be placed. In embodiments, datum element defines the groove at a location overlaying the handgun frame while the handgun is holstered. The finger datum element may provide a high degree of certainty that the user will properly grip the handgun after withdrawal of the handgun from the cavity. In embodiments, the user may use one motion to reach downward to grasp the handgun. In embodiments of the invention, the groove for receiving the user's straightened index finger may be defined by a datum element formed of thermoplastic elastomer material. In embodiments, a finger guiding member is positioned and adapted so that a phalanx of the index finger is in contact with the datum element prior to and as the handgun is being removed from the holster.

(29) A feature and advantage of embodiments is a holster having a finger guide that engages the user's finger and a thumb actuated button that engages the user's thumb. In embodiments, the release actuation mechanism is actuated by the user's thumb rather than the user's index finger.

(30) A feature and advantage of embodiments is a holster having a thumb-actuated button that is not readily accessible or visible to potential attackers. In embodiments, the thumb-actuated button is positioned between two or more protruding portions of the holster body. In embodiments, the thumb-actuated button is positioned between the holster body and a holster mounting plate. In embodiments, the holster mounting plate defined a plurality of slots, the slots being dimensioned and positioned to receive a belt, such as, for example, a police officer's service belt.

(31) A feature and advantage of embodiments is a holster body with an inner polymer layer or liner disposed on an inside surface of an outer polymer shell layer. This arrangement may reduce or eliminate wear and tear on handgun surface finishes. In embodiments, the inner layer or liner is of a softer polymer than the shell portion, the shell portion may be nylon. Portions of the inner layer may be exposed through windows in the shell layer. A feature and advantage of such embodiments is that there is a reduction in noise associated with insertion and withdrawal of the handgun and the finish of the handgun is better protected from scuffing or scratching. In embodiments, the inner softer layer is injection molded first and the nylon layer is molded onto the inner layer. In embodiments, the order may be reversed. In embodiments the dual layer holster body has a tensioning mechanism comprising the holster body with a slit forward from a portion of the holster body that engages the trigger guard of the handgun, the slit extending to a front opening in the forward end of the holster body, the upper portion of the holster body having a peak with two inclined holster body wall portions joined at the peak. A threaded fastener extending through aligned holes forward of the trigger guard receiving region of the holster body may be tightened to pull the opposing sides of the holster body together effecting a clamping on the four corners of the forward holster body and slide.

(32) A feature and advantage of embodiments is a holster that is user convertible between a level two retention level and a level three retention level. In embodiments, a thumb actuated lever that releases a shroud over a rearward opening to the holster cavity also engages and depresses a tab portion on a trigger block lever that rotates the lever to move a trigger guard blocking member out of a blocking position. The thumb actuated lever and shroud may be removed and the trigger block lever with the tab portion is replaced with a lever with a push button thereon. In embodiments, the trigger block lever may be readily removed and replaced with a thumb actuated lever having a thumb pushbutton, the alternate levers may be snapped or seated into a lever receiving region on a proximal side of the holster body, the levers rotatable along a lever axis when seated in the lever receiving region. Each of the alternate levers may be captured and secured in the lever receiving region by a holster mounting portion such as a belt loop or a jacket slot belt loop. This provides the advantage that the user can select and modify her holster for either level two or level three retention rather than buying one level three holster and one level two holster.

(33) A feature and advantage of embodiments is an advantageous method of manufacturing holster utilizes a single mold for the holster body for both a level two holster and a level three holster. The mold having features for a proximal side belt loop mounting region, for example a flat surface with

three holes. Above the mounting region structure (positionally the “upper” holster receives the slide of the handgun), the mold having structure for molding features for a first lever receiving seat and other features associated with a first active handgun retention mechanism, including for example, openings in the proximal holster wall portion forward of the lever receiving seat and rearward of the lever receiving seat such that the molded holster body can receive portions of a first lever of the first active handgun receiving region. The mold also has a second lever receiving region and features for receiving a second active handgun retention mechanism with a shroud pivotally attached at a rear opening of the holster body. In embodiments, mold inserts may be utilized to preclude the molded holster body from having the features for receiving the second retention mechanism. For example, a mold insert will be placed in the mold when openings for attaching the second retention mechanism are desired, those mold inserts removed when the molded holster body is intended for a level two retention. Use of the same mold makes the production more economical resulting in lower prices to the consumer and/or better margins in manufacturing.

(34) A feature and advantage of embodiments is a thumb actuated button of a retaining mechanism is positioned to allow for intuitive and ergonomic release of the handgun from the holster. This allows the user to press his or her thumb against the thumb button with less tension in the muscles and tendons of the thumb and forearm. The thumb button position provides for natural and comfortable movement of the thumb while the forearm is extending in a downward direction.

(35) A feature and advantage of embodiments is a thumb actuated button that can be pivoted with little tension in the muscles and tendons of the forearm and thumb while the forearm is extending in a downward direction and the thumb is contacting the thumb actuated button. In embodiments, the thumb actuated button pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster. In embodiments, the retaining mechanism comprises a lever having a thumb button portion, and the lever pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster. In embodiments, the retaining mechanism comprises a thumb-actuated button that pivots about a first axis and a lever that pivots about a second axis that is parallel to the first axis.

(36) The retention mechanism for the holster may include a lever pivotally supported by the holster body. The lever may comprise an elongate central portion integrally formed with a first forward pivot portion, and the holster body may comprise a second forward pivot portion integrally formed with a wall portion of the holster body. The first forward pivot portion may be mated with the second forward pivot portion so that the lever is pivotally supported by the holster body. In embodiments, one of the forward pivot portions is a pin portion and the other of the forward pivot portions is a C-shaped pin receiving portion. In embodiments, the C-shaped pin receiving portion has a circumferential span less than or equal to 180 degrees. In embodiments, the C-shaped pin receiving portion has a circumferential span greater than 180 degrees. In embodiments, the pin receiving portion may extend 360 degrees. In embodiments, one of the forward pivot portions is a pin portion and the other of the forward pivot portions is a U-shaped pin receiving portion. The lever may also include a first rearward pivot portion integrally formed with the elongate central portion and the holster body may comprise a second rearward pivot portion integrally formed with a wall portion of the holster body. The first rearward pivot portion may be mated with the second rearward pivot portion so that the lever is pivotally supported by the holster body. In embodiments, one of the rearward pivot portions is a pin portion and the other of the rearward pivot portions is a C-shaped pin receiving portion. In embodiments, the C-shaped pin receiving portion has a circumferential span less than or equal to 180 degrees. In embodiments, the C-shaped pin receiving portion has a circumferential span greater than 180 degrees. In embodiments the circumferential span is greater than 185 degrees. In embodiments, the circumferential span is greater than 190 degrees. In embodiments, the circumferential span is 360 degrees. In embodiments, one of the rearward pivot portions is a pin portion and the other of the rearward pivot portions is a U-shaped pin receiving portion. In embodiments the lever may have only one pivot portion and the body only

one cooperating pivot portion. In embodiments, the pivot portion of the lever extends substantially or mostly the length of the lever.

(37) In embodiments, the pin portion is configured as a pintle and the pin receiving portion is configured as a gudgeon. In embodiments, the pintle is unitary with the lever of the retention mechanism and the gudgeon is unitary with the holster body.

(38) A holster in accordance with this detailed description may comprise a user attachment means such as a belt engaging member, a holster body and a lever disposed between the holster body and the user attachment means. The lever may be pivotally supported by the holster body and may be moveable between a handgun trigger guard capture position and a handgun trigger guard release position. In embodiments, the belt engaging member is fixed to the holster body by a plurality of screws. The lever, may be, for example, captured between the user attachment means and the holster body. In embodiments, the lever can be freely separated from the holster body after the user attachment means is removed from the holster body. In embodiments, the user attachment means is a belt engaging member comprising a first belt loop portion defining a first belt passageway. In embodiments, the belt engaging member comprises a second belt loop portion defining a second belt passageway.

(39) In embodiments, the user attachment means is a jacket slot belt loop that has a lower plate portion having a holster attachment region, a mid level portion, and an upper portion with a pair of belt loops. The lower plate portion having apertures for receiving threaded fasteners for attachment of the holster at the holster attachment region. The mid level portion may have a distal side that cooperates with the holster body to capture components of a retention mechanism between the holster body and the mid level portion and thereabove a pair of upright ribs protruding outwardly from the distal side defining a recess therebetween that extends upwardly to the upper portion and is open upwardly. The upper portion having a central column with an upper margin. A pair of belt loop members are displaced distally inward of the central column and displaced laterally from the central column. The belt loop members connect to the central column at upper connecting portions, the central column, the upper connection portions and the belt loop members all unitary with each other. Each belt loop member having a lower end not unitarily joined to the central column but having a closable spacing therefrom. The central column may have protruding portions configured as bosses to cooperate with the lower ends of the belt loop members. Threaded fasteners may be utilized to adjust the spacing between each belt loop member and the central column effecting a clamping action onto a belt, such as a duty belt, whereby the jacket slot belt loop is secured to a user's belt. Optional elastomeric bushings may be utilized between the lower ends of the belt loop members and bosses of the central column. A feature and advantage of such embodiments is that the level of clamping of the jacket slot belt loop and holster are readily adjustable. Moreover, the lateral offset of the belt loop members from the central column provides stability for the mounting system and holster as the length of the engagement of the jacket slot belt loop with the holster is extended.

(40) The ribs may provide structural strengthening of the plate portion as well as providing access inhibiting structure to prevent access by others from front of the holster or the back side of the holster, as the holster is worn. The recess opening upwardly providing access to the holster wearer as well as guide structure to easily and non-visually guide the users thumb to the proper location both to actuate a thumb release button and for the master grip.

(41) A holster for receiving a handgun having a trigger guard, comprises a holster body supporting a retention mechanism. In embodiments, the holster body has a plurality of holster wall portions defining a handgun holding cavity extending along a handgun receiving and withdrawal axis. In embodiments, the retention mechanism comprises a lever pivotally supported by the holster body. In embodiments, the lever with a pivoting connection length of the lever extending more than half of the length of the lever, providing stability of the lever and holster body interface. In embodiments, the lever comprising an elongate central portion integrally formed with a first

forward pivot portion and the holster body comprising a second forward pivot portion integrally formed with one of the holster wall portions, the first forward pivot portion mating with the second forward pivot portion. In embodiments, the lever further comprises a first rearward pivot portion integrally formed with the elongate central portion and the holster body comprising a second rearward pivot portion integrally formed with one of the holster wall portions, the first rearward pivot portion mating with the second rearward pivot portion. In embodiments, one of the forward pivot portions is a pin portion and the other of the forward pivot portions is a C-shaped pin receiving portion. In embodiments, one of the rearward pivot portions is a pin portion and the other of the rearward pivot portions is a C-shaped pin receiving portion. In embodiments, the lever is pivotally attached to the holster body and is moveable between a handgun trigger guard capture position and a handgun trigger guard release position.

(42) In embodiments, a holster body has exterior bosses for receiving either a handgun button guard for a holster with a single active retention mechanism, or for receiving components of a second retention mechanism. A feature and advantage is the same holster body can be utilized for two different holsters, one with a single active retention mechanism and one with two active retention mechanisms.

(43) In embodiments, a holster body has an inner liner supported by an outer layer, the inner liner of a polymer softer than the polymer of the outer layer, the holster body defining a pair of opposing V-shaped portions for supporting the forward portion of the handgun, the V-shaped portions having an upper inverted V portion and a lower V portion, the V-shaped portions for engaging with the slide and body corners of the forward portion of the handgun, the lower V-shaped portion having an tensioning screw for adjusting the spacing of opposing legs of the lower V-shaped portion.

(44) In embodiments, a forward holster body has a rhombus or diamond shape, with an adjustable gap at the bottom of where four corners of the forward portion of a handgun engages intermediate portions of each side of the diamond shape. An adjustment screw at the bottom of diamond causes contraction or expansion of the diamond shape allowing adjustment of the engagement and gripping of the holster on the handgun. A softer inner layer of the holster body enhances the gripping function.

(45) In embodiments, a jacket slot belt loop attaches to a proximal wall portion of a holster, the holster having a active retention mechanism on the proximal wall portion of the holster, the jacket slot belt loop capturing components of the active retention mechanism between the proximal wall portion of the holster body and the jacket slot belt loop.

(46) In embodiments, the U-shaped shroud of the holster is part of a hood assembly. In embodiments, a bracket of the hood assembly includes a lug portion and the thumb receiving tab includes a protrusion portion that engages the lug portion of the hood assembly while the U-shaped shroud is in the handgun retaining position. In embodiments, the holster further includes a hood spring that biases the U-shaped shroud to pivot toward the release position. In embodiments, the hood spring biases the lug portion of the hood assembly against the protrusion portion of the thumb receiving tab while the U-shaped shroud is in the handgun retaining position. In embodiments, rotation of the thumb receiving tab through a first range of rotary motion causes the protrusion portion of the thumb receiving tab to disengage from the lug portion of the hood assembly. In embodiments, rotation of the thumb receiving tab through a second range of rotary motion causes the thumb receiving tab to engage the tab receiving portion of the lever causing at least a portion of the trigger guard retaining portion of the lever to be withdrawn from the trigger guard receiving portion of the holster body.

(47) In embodiments, the holster includes the hood spring, a lever spring and a tab spring. In embodiments, each spring comprises a length of wire, the wire of the spring forming a first leg, a second leg and a coil disposed between the first leg and the second leg. In embodiments, the first leg of the hood spring is fixed relative to the holster body and the second leg of the hood spring is seated against the hood assembly so that the U-shaped shroud is biased to pivot toward the release

position. In embodiments, the lever spring is positioned and adapted to bias the lever to rotate toward the handgun trigger guard capture position. In embodiments, the first leg of the lever spring is seated against to the holster body and the second leg of the lever spring is seated against the lever. In embodiments, the tab spring is positioned and adapted to bias the thumb receiving tab to rotate in a direction that moves a distal end of the thumb receiving tab away from the holster body. In embodiments, the first leg of the tab spring is seated against to the holster body and the second leg of the tab spring is seated against the thumb receiving tab.

(48) The above summary is not intended to describe each illustrated embodiment or every implementation of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings included in the present application are incorporated into, and form part of, the specification. They illustrate embodiments of the present disclosure and, along with the description, serve to explain the principles of the disclosure. The drawings are only illustrative of certain embodiments and do not limit the disclosure.

(2) FIG. 1 is a perspective view showing a handgun and a holster in accordance with the detailed description.

(3) FIG. 2 is a perspective view showing a handgun and a holster in accordance with the detailed description.

(4) FIG. 3A is an exploded perspective view showing a holster body and a lever.

(5) FIG. 3B is an exploded perspective view showing a belt receiving member, a holster body and a lever.

(6) FIG. 3C is an exploded perspective view showing a belt receiving member, a holster body and a lever.

(7) FIG. 4 is an exploded perspective view showing a holster body and a lever.

(8) FIG. 5 is a perspective view of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIG. 5.

(9) FIG. 6 is a perspective view of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIG. 6.

(10) FIG. 7 is a perspective view of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIG. 7.

(11) FIGS. 8A and 8B are plan views of an assembly including a lever and a handgun having a trigger guard. The handgun is cross-sectioned for purposes of illustration in FIGS. 8A and 8B.

(12) FIG. 9 is a cross-sectioned perspective view of an assembly including a holster and a handgun having a trigger guard. The holster and the handgun are cross-sectioned for purposes of illustration in FIG. 9.

(13) FIG. 10 is a plan view of an assembly including a holster and a handgun having a trigger guard. The holster and the handgun are cross-sectioned for purposes of illustration in FIG. 10.

(14) FIGS. 11A-11C are stylized cross-sectional views each showing a pin and a pin receiving portion.

(15) FIG. 12 is a stylized cross-sectional view showing a pin and a pin receiving portion.

(16) FIG. 13 is a perspective view showing a pin and a pin receiving portion.

(17) FIG. 14 is a perspective view showing a handgun and a holster in accordance with the detailed description.

(18) FIG. 15 is an exploded perspective view showing a holster body and two active handgun retention mechanisms operated by a single thumb button.

(19) FIG. 16 is a perspective view showing retention mechanisms seen in the exploded perspective

view of FIG. 15.

(20) FIG. 17 is an exploded perspective view further illustrating some of the elements seen in the sub-assembly of FIG. 16.

(21) FIG. 18A is a perspective view of components of handgun retention mechanisms.

(22) FIG. 18B is an exploded view of the components of FIG. 18A.

(23) FIG. 19 is a further exploded view of components of the handgun retention mechanisms including a housing for securing the pivoting thumb button.

(24) FIG. 20 is a further exploded view of components of the handgun retention mechanisms taken from the side opposite that of FIG. 19.

(25) FIG. 21 is an exploded view of an embodiment of a holster with a with thumb button operating a single retention mechanism and a thumb button guard.

(26) FIG. 22 is a plan view of a holster with a holstered handgun and a user's hand about the grasp the handgun.

(27) FIG. 23 is a plan view of a holster with a holstered handgun and a user's hand engaging the handgun.

(28) FIG. 24 is a plan view of a holster with a holstered handgun and a user's hand engaging the handgun.

(29) FIG. 25 is a perspective view showing a holster including a hood spring, a lever spring, and a tab spring. For purposes of illustration and explanation, the hood spring, the lever spring, and the tab spring are each illustrated a second time at a location spaced away from the holster.

(30) FIG. 26 is a perspective view showing an assembly including a lever, a lever spring, a tab, and a tab spring. For purposes of illustration and explanation, the lever spring and the tab spring are each illustrated a second time at a location spaced away from the assembly.

(31) FIG. 27 is a perspective view of an assembly including a lever, a lever spring, a tab, and a tab spring. For purposes of illustration and explanation, the lever spring and the tab spring are each illustrated a second time at a location spaced away from the assembly.

(32) FIGS. 28A and 28B are perspective views showing an assembly including a hood assembly and a thumb receiving tab.

(33) FIGS. 29A and 28B are perspective views showing an assembly including a hood assembly and a thumb receiving tab.

(34) FIG. 30A is a plan view showing a hood assembly part having a lug portion and a thumb receiving tab having a protrusion portion that selectively engages the lug portion of the hood assembly part. FIG. 31A is a perspective view of the assembly shown in FIG. 30A. In the embodiments of FIGS. 30A and 31A, the hood assembly part is in a handgun retaining position.

(35) FIG. 30B is a plan view showing a hood assembly part having a lug portion and a thumb receiving tab having a protrusion portion that selectively engages the lug portion of the hood assembly part. FIG. 31B is a perspective view of the assembly shown in FIG. 30B. In the embodiments of FIGS. 30B and 31B, the hood assembly part is in a handgun releasing position.

(36) FIG. 32A is a perspective view showing a holster in accordance with the detailed description.

(37) FIG. 32B is a perspective view showing selected parts from the holster shown in FIG. 32A. The parts shown in FIG. 32B include a U-shaped shroud, a bracket and a finger receiving tab.

(38) FIG. 32C is a perspective view further illustrating the parts shown in FIG. 32B. In FIG. 32C, the U-shaped shroud, the bracket and the finger receiving tab are shown from a different viewpoint.

(39) FIGS. 33A and 33B are two perspective views showing a holster including a U-shaped shroud. In the embodiment of FIG. 33A, the U-shaped shroud is in a handgun retaining position. In the embodiment of FIG. 33B, the U-shaped shroud is in a release position.

(40) FIGS. 34A and 34B are perspective views of a belt engaging member.

(41) FIG. 35A is a front view of the belt engaging member shown in FIGS. 34A and 34B.

(42) FIG. 35B is a right side view of the belt engaging member shown in FIGS. 34A and 34B.

(43) FIG. 35C is a top view of the belt engaging member shown in FIGS. 34A and 34B.

(44) FIG. 35D is a rearview of the belt engaging member shown in FIGS. 34A and 34B.

(45) FIG. 35E is a left side view of the belt engaging member shown in FIGS. 34A and 34B.

(46) FIG. 35F is a bottom view of the belt engaging member shown in FIGS. 34A and 34B.

(47) FIGS. 36A and 36B are two perspective views showing a holster having a first handgun retention mechanism including a pivotable lever and a second handgun retention mechanism including a U-shaped shroud. In the embodiment of FIGS. 36A and 36B, the U-shaped shroud is in a handgun retaining position.

(48) FIG. 37A is a front view of the holster shown in FIGS. 36A and 36B.

(49) FIG. 37B is a right side view of the holster shown in FIGS. 36A and 36B.

(50) FIG. 37C is a top view of the holster shown in FIGS. 36A and 36B.

(51) FIG. 37D is a rear view of the holster shown in FIGS. 36A and 36B.

(52) FIG. 37E is a left side view of the holster shown in FIGS. 36A and 36B.

(53) FIG. 37F is a bottom view of the holster shown in FIGS. 36A and 36B.

(54) FIGS. 38A and 38B are two perspective views showing a holster having a handgun retention mechanism including a pivotable lever.

(55) FIG. 39A is a front view of the holster shown in FIGS. 38A and 38B.

(56) FIG. 39B is a right side view of the holster shown in FIGS. 38A and 38B.

(57) FIG. 39C is a top view of the holster shown in FIGS. 38A and 38B.

(58) FIG. 39D is a rear view of the holster shown in FIGS. 38A and 38B.

(59) FIG. 39E is a left side view of the holster shown in FIGS. 38A and 38B.

(60) FIG. 39F is a bottom view of the holster shown in FIGS. 38A and 38B.

(61) FIG. 40 is an exploded perspective view showing a belt receiving member, a holster body and a lever.

(62) FIG. 41 is an exploded perspective view showing a holster body and a lever.

(63) FIG. 42 is an exploded perspective view showing a belt receiving member, a holster body and two locking mechanisms.

(64) FIG. 43 is an exploded perspective view showing a holster body and two locking mechanisms.

(65) FIG. 44 is a perspective view showing the belt receiving member, the holster body and the lever shown in FIG. 40.

(66) FIG. 45 is a top view showing the belt receiving member, the holster body and the lever shown in FIG. 40.

(67) FIG. 46 is a perspective view showing the belt receiving member, the holster body and the locking mechanisms shown in FIG. 42.

(68) FIG. 47 is a top view showing the belt receiving member, the holster body and the locking mechanisms shown in FIG. 42.

(69) FIGS. 48A and 48B are perspective views of a belt engaging member.

(70) FIG. 49A is a front view of the belt engaging member shown in FIGS. 48A and 48B.

(71) FIG. 49B is a right side view of the belt engaging member shown in FIGS. 48A and 48B.

(72) FIG. 49C is a top view of the belt engaging member shown in FIGS. 48A and 48B.

(73) FIG. 49D is a rear view of the belt engaging member shown in FIGS. 48A and 48B.

(74) FIG. 49E is a left side view of the belt engaging member shown in FIGS. 48A and 48B.

(75) FIG. 49F is a bottom view of the belt engaging member shown in FIGS. 48A and 48B.

(76) FIGS. 50A and 50B are two perspective views showing a holster having a first handgun retention mechanism including a pivotable lever and a second handgun retention mechanism including a U-shaped shroud. In the embodiment of FIGS. 50A and 50B, the U-shaped shroud is in a handgun retaining position.

(77) FIG. 51A is a front view of the holster shown in FIGS. 50A and 50B.

(78) FIG. 51B is a right side view of the holster shown in FIGS. 50A and 50B.

(79) FIG. 51C is a top view of the holster shown in FIGS. 50A and 50B.

(80) FIG. 51D is a rear view of the holster shown in FIGS. 50A and 50B.

- (81) FIG. 51E is a left side view of the holster shown in FIGS. 50A and 50B.
- (82) FIG. 51F is a bottom view of the holster shown in FIGS. 50A and 50B.
- (83) FIGS. 52A and 52B are two perspective views showing a holster having a handgun retention mechanism including a pivotable lever.
- (84) FIG. 53A is a front view of the holster shown in FIGS. 52A and 52B.
- (85) FIG. 53B is a right side view of the holster shown in FIGS. 52A and 52B.
- (86) FIG. 53C is a top view of the holster shown in FIGS. 52A and 52B.
- (87) FIG. 53D is a rear view of the holster shown in FIGS. 52A and 52B.
- (88) FIG. 53E is a left side view of the holster shown in FIGS. 52A and 52B.
- (89) FIG. 53F is a bottom view of the holster shown in FIGS. 52A and 52B.

(90) While the embodiments of the disclosure are amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

(91) Referring to FIGS. 1-4, perspective views of a handgun **80** and a holster **100** are shown. FIG. 1 shows the handgun **80** withdrawn from the holster **100** and FIG. 2 shows the handgun inserted into the holster **100**. The handgun being conventional and having a forward end **82**, a handgun body **84**, a grip **86** at a rearward end **87** of the handgun **80**, a trigger guard **90**, a back strap **92** with a backstrap recess **94**, a slide **98** positioned above the handgun body, and a rail **99** positioned below the slide. In embodiments, the holster **100** for receiving and withdrawing the handgun having a trigger guard **90** comprises a polymer holster body **102**, an elongate polymer pivoting lever **104** that is part of a first handgun retention mechanism **106** that is thumb actuated. The pivoting lever has an actuation tab **107** configured as by a thumb receiving button that is disposed in an opening **105** in the holster body, actuation of the thumb receiving button moves a trigger guard block portion **128** of the lever between retention and release positions. The retention mechanism **106** has two distinct separated pivoting connections **109** each formed from a pair of cooperating connector pivot portions **110**, one of each pair unitary with the holster body and the other of each pair unitary with the lever. In embodiments, the pivot portions **110** of the lever **104** and pivot portions **110** of the holster body **102** are coupled using a simple assembly process with no separate hinge pin.

(92) The holster body has three unitary bosses **111** on a proximal wall portion **113** for attachment to a plate portion of a user attachment means, such as a belt engaging member. See FIGS. 12 and 13. The holster having a proximal side and the holster body having a proximal side.

(93) Continuing to referring to FIGS. 1-4, in embodiments, the holster body **102** has a plurality of holster wall portions defining a cavity **108** extending along a handgun receiving and withdrawal axis **110H**. In embodiments, the retention mechanism **106** comprises a lever **104** pivotally supported by the holster body **102** and retained by holster attachment plate portion **165** or by other means. The lever **104** may comprise an elongate central portion **124** integrally formed with a first forward pivot portion **112** and the holster body **102** may comprises a second forward pivot portion **114** integrally formed with one of the holster wall portions. The first forward pivot portion **112** may mate with the second forward pivot portion **114** to form a forward pivoting connection **109**. In embodiments, the lever **104** further comprises a first rearward pivot portion **116** integrally formed with the elongate central portion **124** of the lever **104** and the holster body **102** comprises a second rearward pivot portion **118** integrally formed with one of the holster wall portions. The first rearward pivot portion **116** may mate with the second rearward pivot portion **118** to form a rearward pivoting connection **109**. In embodiments all pivot portions of the retention mechanism are axially aligned and co-axial when assembled.

(94) Continuing to referring to FIGS. 1-4, in embodiments, is a retention mechanism having a lever that can be pivoted with little tension in the muscles and tendons of the forearm and thumb while

the forearm is extending in a downward direction and the thumb is contacting the a thumb button portion of the lever. In embodiments, the retaining mechanism comprises a lever having a thumb button portion, and the lever pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster. In embodiments, the retaining mechanism comprises a thumb-actuated button that pivots about a first axis and a lever that pivots about a second axis that is parallel to the first axis.

(95) Referring to FIGS. 5-8B, in embodiments, the lever **104** of the retention mechanism **106** has an elongate central portion **124**, a thumb receiving portion **107**, configured as a thumb button, and a blocking portion **108**. The thumb receiving portion of the lever comprising a first arm **130** extending away from the elongate central portion **124** in a first direction, the elongate central portion **124** and the first arm **130** cooperating to form a first L-shaped portion **132**. The blocking portion **128** of the lever **104** comprises a second arm **134** extending away from the elongate central portion **124** in a second direction opposite the first direction, the elongate central portion **124** and the second arm **134** cooperating to form a second L-shaped portion **136**.

(96) Referring to FIGS. 6 and 7, the lever **104** is elongate in the direction of the axis X and has a lever length L1. The pivot portions **112**, **116** have a separation distance D1 and have a pivoting connection length L2. The pivoting connection length L2 to the overall lever length L1 is in embodiments greater than 0.70 or 70%; in embodiments, greater than 0.60 or 60%; in embodiments greater than 0.80 or 80%.

(97) Referring to FIGS. 1 and 2, a forward or handgun insertion direction Z and a rearward or handgun withdrawal direction -Z are illustrated using arrows labeled "Z" and "-Z," respectively. An upward direction Y and a downward direction -Y are illustrated using arrows labeled "Y" and "-Y," respectively. A direction X extending away from the user's body and the user attachment side of the holster is illustrated using an arrow labeled "X." A direction -X extending toward the user's body and the user attachment side of the holster is illustrated using an arrow labeled "-X." The directions illustrated using these arrows may be conceptualized, by way of example and not limitation, from the point of view of a user who is wearing a holster hung from a service belt and inserting a handgun into the holster. The directions illustrated using these arrows may also be conceptualized, by way of example and not limitation, from the point of view of a user holding a handgun in a normal firing position and viewing the gunsights of the handgun. The directions illustrated using these arrows may be applied to the apparatus shown and discussed throughout this application. In embodiments, the Z direction and the -Z direction are both generally orthogonal to the XY plane defined by the X direction and the Y direction. In embodiments, the X direction and the -X direction are both generally orthogonal to the ZY plane defined by the Z direction and the -Z direction. In embodiments, the Y direction and the -Y direction are both generally orthogonal to the ZX plane defined by the Z direction and the X direction. Various direction-indicating terms are used herein as a convenient way to discuss the objects shown in the figures. It will be appreciated that many direction indicating terms are related to the instant orientation of the object being described. It will also be appreciated that the objects described herein may assume various orientations without deviating from the spirit and scope of this detailed description. Accordingly, direction-indicating terms such as "upwardly," "downwardly," "forwardly," "rearwardly," etc. should not be interpreted to limit the scope of the invention recited in the attached claims.

(98) Referring to FIG. 9, a cross-sectioned perspective view of a holster body **102** is shown. In the example embodiment of FIG. 9, the holster body **102** is formed from two polymers, one overmolded onto the other, providing an interior layer **138** and an exterior layer **140**. In embodiments, the interior layer **138** is softer than the exterior layer **140** minimizing any wear, damage or markings on the exterior surface of the handgun that is received in the cavity **108** defined by the holster. In embodiments, the interior layer **138** and the exterior layer **140** are combined using an assembly process. In embodiments, the interior layer **138** comprises a thermoplastic elastomer (TPE) and the exterior layer **140** a thermoplastic material. In embodiments,

the interior layer **138** comprises Hytrel® polymer, available from DuPont, and the exterior layer **140** comprises nylon. The exterior layer being harder and stiffer than the interior layer.

(99) Referring to FIG. **10**, a cross-sectional view of a holster body **102** is shown. In embodiments, the holster body **102** is formed to provide a handgun fitting function that can be adjusted by rotating a tensioning screw **142**. The shape of the forward portion of the handgun can be conceptualized as a four cornered geometric figure, roughly a rectangle, and the shape of the cavity **108** defined by the holster body **102** can be configured to make point contact with the four corners of the figure or rectangle. In embodiments, the shape of the cavity **108** defined by the holster body **102** can be conceptualized as two opposing V-shapes, an upper V-shape **148** and a lower V-shape **146**. The upper V-shape **148** and the lower V-shape **146** are shown using dashed lines in FIG. **10**. The two legs of the upper V-shape **148** and the two legs of the lower V-shape **146** may each contact one corner of the figure or rectangle. In embodiments, the lower V-shape **146** defines a slot **144** near the lower end of the lower V-shape **146**. A tensioning screw **142** is positioned to selectively decrease the angle between the two legs of the lower V-shape **146**. As the angle between the two legs of the lower V-shape **146** decreases, the two legs of the lower V-shape **146** apply upwardly directed component forces to two lower corners **150** of the rectangle. The forces applied to the lower corners **150** of the rectangle urge the two upper corners **152** of the rectangle against the two legs of the upper V-shape **148**.

(100) The components herein may be formed of thermoplastic polymers using an injection molding process.

(101) Referring to FIGS. **3**, **4** and **11A-11C**, a retention mechanism **106** for a holster **100** may include a lever pivotally supported by the holster body **102**. The lever **104** may comprise an elongate central portion **124** integrally formed with a first forward pivot portion **112**, and the holster body **102** may comprise a second forward pivot portion **114** integrally formed with a wall portion of the holster body **102**. The first forward pivot portion **112** may be mated with the second forward pivot portion so that the lever **104** is pivotally supported by the holster body **102**. In embodiments, one of the forward pivot portions **112**, **114** is a pin portion **154** and the other of the forward pivot portions **112**, **114** is a C-shaped pin receiving portion **156**. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span less than or equal to 180 degrees. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span greater than 180 degrees; in embodiments greater than 185°; and in embodiments greater than 190°. In embodiments, one of the forward pivot portions **112**, **114** is a pin portion **154** and the other of the forward pivot portions **112**, **114** is a U-shaped pin receiving portion **158**.

(102) With continuing reference to FIGS. **3**, **4** and **11A-11C**, the lever **104** may further comprise a first rearward pivot portion **116** integrally formed with the elongate central portion **124** and the holster body **102** may comprise a second rearward pivot portion **118** integrally formed with a wall portion of the holster body **102**. The first rearward pivot portion **116** may be mated with the second rearward pivot portion so that the lever **104** is pivotally supported by the holster body **102**. In embodiments, one of the rearward pivot portions **116**, **118** is a pin portion **154** and the other of the rearward pivot portions **116**, **118** is a C-shaped pin receiving portion **156**. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span less than or equal to 180 degrees. In embodiments, the C-shaped pin receiving portion **156** has a circumferential span greater than 180 degrees; in embodiments greater than 185°; and in embodiments greater than 190°. In embodiments, one of the rearward pivot portions **116**, **118** is a pin portion **154** and the other of the rearward pivot portions **116**, **118** is a U-shaped pin receiving portion **158**.

(103) Referring to FIGS. **3B** and **3C**, a holster **100** in accordance with this detailed description may comprise a holster attachment plate portion **165** that is part of or all of a user attachment means for the holster which may be a belt engaging member. The plate portion **165** may retain the lever **104** in a pivoting connection arrangement with the holster body **102**. The lever **104** may be pivotally supported by the holster body **102** and may be moveable between a handgun trigger guard capture

position and a handgun trigger guard release position. In embodiments, the plate portion **165** is fixed to the holster body **102** by a plurality of screws **162**. The lever **104**, may be, for example, captured between the belt engaging member **160**, or other user attachment means, and the holster body **102**. In embodiments, the lever **104** can be freely separated from the holster body **102** after the belt engaging member **160** is removed from the holster body **102**. In embodiments, the belt engaging member **160** comprises a first belt loop portion **164** defining a first passageway **166** to receive a belt and a second belt loop portion **168** defining a second passageway **170** for receiving the belt.

(104) The user attachment means **160** may comprise various holster supporting devices without deviating from the spirit and scope of this detailed description. Examples of holster supporting devices that may be suitable in some applications are disclosed in the following United States Patents all of which are hereby incorporated by reference herein: U.S. D653848, U.S. D567707, U.S. D508318, U.S. Pat. Nos. 9,134,093, 8,783,532, 8,517,234, 8,469,245, 8,297,562, 8,251,266, 8,235,263, 7,866,515, 7,320,420, 9,423,210, 9,664,480, 9,841,255, 9,222,751, 8,544,706, 8,215,525, 8,100,304, 7,971,762, 7,922,050, 7,690,541, 6,478,202, 6,189,751, and 5,467,909.

(105) Referring to FIGS. **14-17**, in embodiments, a holster **100** has a first handgun retention mechanism **106'** that engages the trigger guard as previously described with respect to FIGS. **1-13**, and a second handgun retention mechanism **106'** with a pivoting lever **104'**. A dual actuation member **179** with a thumb button **180** actuates both the first handgun retention mechanism **106'** and the second handgun retention mechanism **174**. In embodiments, each handgun retention mechanism selectively prevents the handgun **80** from being withdrawn from the cavity **108** defined by the wall portions **109** of the holster body **102**.

(106) The second handgun retention mechanism **106'** comprises a slide retention member configured as a U-shaped shroud **176** that is pivotally supported by the holster body **102** so that the U-shaped shroud **176** selectively pivots between a handgun retaining position in which the U-shaped shroud **176** extends across a portion of a rearward opening **178** of the cavity **108** and a release position in which the U-shaped shroud **176** extends forward of the rearward opening **178** so that a withdrawal path of the handgun **80** is unobstructed. In embodiments, the U-shaped shroud **176** pivots about a shroud pivot axis **177** that extends laterally and is perpendicular to the handgun insertion and withdrawal axis.

(107) Referring to FIGS. **15-20**, in embodiments, the first handgun retention mechanism **106'** comprises a lever **104'** pivotally supported by the holster body **102**. The lever **104'** may be similarly configured to the lever **106** of FIGS. **1-13**. In embodiments, the lever **104'** has an elongate central portion **124**, a depressible actuation portion **186**, and a trigger guard blocking portion **128**. The depressible actuation portion **186** of the lever **104** comprises a first arm **130** extending away from the elongate central portion **124** in a first direction so that the elongate central portion **124** and the first arm **130** cooperate to form a first L-shape **182**. The trigger guard blocking portion **128** of the lever **104'** comprises a second arm **134** extending away from the elongate central portion **124** in a second direction opposite the first direction so that the elongate central portion **124** and the second arm **134** cooperate to form a second L-shape **184**. In embodiments, the lever **104'** is pivotally supported by the holster body so that the lever **104'** pivots about a lever axis **110** that extends in upward and downward directions as the holster is worn or forward and rearward with respect to the handgun. In embodiments, the lever pivots between a handgun trigger guard blocking or capture position and a handgun trigger guard non-blocking or release position. The trigger guard blocking portion **128** of the lever **104** extends into a trigger guard portion of the holster body **102** when the lever **104'** is disposed in the handgun trigger guard capture position.

(108) Continuing to referring to FIGS. **15-20**, a feature and benefit of embodiments is a retention mechanism having a lever that can be pivoted with little tension in the muscles and tendons of the forearm and thumb while the forearm is extending in a downward direction and the thumb is used to rotate the lever. In embodiments, the retaining mechanism comprises a thumb-actuated button

that pivots about a first axis and a lever that pivots about a second axis that is parallel to the first axis. In embodiments, the lever pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster. In embodiments, the thumb-actuated button pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster.

(109) Referring to FIGS. **16-20**, the second handgun retention mechanism is illustrated. The slide blocking member configured as a shroud **176** is part of a hood assembly **188**. Fasteners **197**, **198** pivotally secure the U-shaped shroud, second handgun retention mechanism housing **201**, bracket **190** with slot **144**, to the holster body. Fastener **199** secures the opposite side of the shroud to the holster body. The second handgun retention mechanism **174** utilizes the retention mechanism housing **201** to secure components of the retention mechanism in place. The housing has two cooperating components, and inward member **202** and an outward member **204**, that may be arranged in a clamshell-like fashion that captures the leaf spring **194** and a pivoting shaft **206** of the dual actuation member **179** within the housing **201**. In embodiments, the holster includes a spring **196** that biases the U-shaped shroud **176** to pivot toward the release position. In embodiments, the holster **100** includes the leaf spring **194** that is received in the slot **144** of the bracket **190** when the U-shaped shroud **176** is in the handgun retaining position. The base **195** of the leaf spring **194** is fixed with respect to the holster body and the tip **196** of the leaf spring **194** can flex inwardly and outwardly. The bracket **190** is attached to the U-shaped shroud **176** such that as the leaf spring **194** keeps the bracket from rotating forwardly it also keeps the shroud from rotating forwardly. In embodiments, rotation of the thumb receiving tab or button **180** through a first range of rotary motion causes deflection of the leaf spring **194** so that a distal portion of the leaf spring **194** moves outside of the slot **144** allowing the bracket **190** and also the shroud **176** to rotate forwardly. In embodiments, rotation of the thumb button **180** through a second range of rotary motion causes the thumb button **180** to engage the depressible actuation portion **186** of the lever **104** causing at least part of the trigger guard blocking portion **128** of the lever to be withdrawn from the trigger guard receiving portion of the holster body **102**. In this embodiment, the thumb receiving portion of the lever **104** of the embodiment of FIGS. **1-4** has been replaced with an actuation portion or depressible portion that is depressed by the inner side of the thumb receiving tab **180**.

(110) The first and second active handgun retention mechanisms may be arranged such that the second and first mechanisms can actuate sequentially or simultaneously, in embodiments.

(111) Referring to FIG. **21**, another embodiment of a holster **300** comprising a holster body **306** similar to that of FIGS. **14** and **15** with a top or forward wall portion that mostly covers the slide of a handgun holstered therein, a pair of bosses **310** that can receive components of a second active handgun retention mechanism or, as illustrated in FIG. **21**, a removable thumb button guard **312** secured to the bosses with fasteners **316**. The lever **104** may be configured as discussed previously with respect to FIGS. **1-9** and has a thumb button **180**. The plate portion **165** of a belt engaging member may capture the pivoting lever onto the holster body.

(112) Referring to FIGS. **22-24**, the sequence of gripping and drawing a handgun **80** from a holster **100** having an optimally positioned thumb button **107** is illustrated. The handgrip of the handgun **80** has a backstrap **322** region with a backstrap recess **324** that receives the user's hand **340** initially by the webbing **342** of the hand contacting the recess **324**. The user will then wrap his middle, ring, and pinky fingers **350**, **351**, **352** around the grip as shown in FIG. **23** and will have his forefinger **353** in a longitudinal recess for said finger on the side of the holster opposite the thumb button. The thumb **354** will naturally be positioned at the thumb button **107** such that a normal grasping action of the grip, consistent with the "master grip" will depress the thumb button **107** and actuate the retention mechanism **104**. Referring to FIGS. **22-25**, **36A**, **36B**, **38A**, **38B** and **40-43**, the master grip and ergonomics are further facilitated in embodiments where the thumb button **180** is positioned with no holster body directly rearward of the rearward margin of the thumb button **180**, as well as no holster body portions below the thumb button **180**. In use, as the handgun **80** is gripped prior to actuation of the thumb button **180**, the user's thumb is in the natural gripping

position at the surface of the thumb button **180**, the thumb button **180** is depressed, coincident with grasping the handgun **80**, and as the handgun **80** is withdrawn, the user's thumb slides off of the thumb button **180** into immediate contact with the handgun **80**. That is, there is no engagement with the exterior surface of the holster body **102** by the user's thumb either during actuation of the release of the retention mechanism or as the handgun **80** is withdrawn. In that the thumb button **180** is positioned in close proximity to the surface of the handgun **80**, the thumb movement inward as the thumb transitions from engagement with the thumb button **180** to engagement with the body of the handgun **80** is minimal.

(113) Referring to FIGS. **25-27C**, in embodiments, the holster includes the hood spring **196**, a lever spring **198** and a tab spring **200**. In embodiments, each spring comprises a length of wire **206**, the wire **206** of the spring forming a first leg **208**, a second leg **210** and a coil **212** disposed between the first leg **208** and the second leg **210**. In embodiments, the first leg **208** of the hood spring **196** is fixed relative to the holster body **102** and the second leg **210** of the hood spring **196** is seated against the hood assembly **188** so that the U-shaped shroud **176** is biased to pivot toward the release position. In embodiments, the lever spring **198** is positioned and adapted to bias the lever **104** to rotate toward the handgun trigger guard capture position. In embodiments, the first leg **208** of the lever spring **198** is seated against to the holster body **102** and the second leg **210** of the lever spring **198** is seated against the lever **104**. In embodiments, the tab spring **200** is positioned and adapted to bias the thumb receiving tab **180** to rotate in a direction that moves a distal end of the thumb receiving tab **180** away from the holster body **102**. In embodiments, the first leg **208** of the tab spring **200** is seated against to the holster body **102** and the second leg **210** of the tab spring **200** is seated against the thumb receiving tab **180**.

(114) Referring to FIGS. **28A-31B**, in embodiments, the U-shaped shroud **176** is part of a hood assembly **188**. In embodiments, a bracket **190** of the hood assembly **188** includes a lug portion **214** and the thumb receiving tab **180** includes a protrusion portion **216** that engages the lug portion **214** of the hood assembly **188** while the U-shaped shroud **176** is in the handgun retaining position. In embodiments, the holster **100** further includes a hood spring **196** that biases the U-shaped shroud **176** to pivot toward the release position. In embodiments, the hood spring **196** biases the lug portion **214** of the hood assembly **188** against the protrusion portion **216** of the thumb receiving tab **180** while the U-shaped shroud **176** is in the handgun retaining position. In embodiments, rotation of the thumb receiving tab **180** through a first range of rotary motion causes the protrusion portion **216** of the thumb receiving tab **180** to disengage from the lug portion **214** of the hood assembly **188**. In embodiments, rotation of the thumb receiving tab **180** through a second range of rotary motion causes the thumb receiving tab **180** to engage the tab receiving portion of the lever **104** causing at least a portion of the trigger guard retaining portion of the lever **104** to be withdrawn from the trigger guard receiving portion of the holster body **102**.

(115) Referring to FIGS. **33A** and **33B**, a feature and benefit of embodiments is a retention mechanism having a lever **104** that can be pivoted with little tension in the muscles and tendons of the forearm and thumb while the forearm is extending in a downward direction and the thumb is contacting a thumb receiving tab **180**. In embodiments, the retaining mechanism comprises a thumb receiving tab **180** that pivots about a first axis and a lever **104** that pivots about a second axis that is parallel to the first axis. In embodiments, the lever **104** pivots about an axis that is parallel to a handgun insertion and withdrawal axis **110H** of the holster **100**. In embodiments, the thumb receiving tab **180** pivots about an axis that is parallel to a handgun insertion and withdrawal axis of the holster **100**. FIGS. **33A** and **33B** are two perspective views showing a holster including a U-shaped shroud. In the embodiment of FIG. **33A**, the U-shaped shroud is in a handgun retaining position. In the embodiment of FIG. **33B**, the U-shaped shroud is in a release position.

(116) Referring to FIGS. **1-47**, in embodiments, an exteriorly worn holster **100** for receiving a handgun **80** has a holster body **102** having a plurality of unitary holster wall portions defining a handgun receiving cavity **108** and a rearward opening **88**. In embodiments, the handgun has a

handgun body **84**, a handgrip **86**, a slide **98**, and a trigger guard **90**. In embodiments, the holster body **102** has a handgun receiving and withdrawal axis **110H**. In embodiments, the plurality of holster wall portions comprise a user proximal wall portion **113** and a user distal wall portion **115**, the proximal wall portion **113** having thumb button recess **218** with a thumb button recess edge portion **220** defining the thumb button recess **218**. In embodiments, the thumb button recess **218** extends forwardly from the rearward opening **88**.

(117) In embodiments, a first user actuatable handgun retention mechanism **106** of the holster **100** is located at the proximal wall portion **113** for selectively preventing the handgun **80**, when in the cavity **108**, from being withdrawn from the cavity **108**. In embodiments, the first handgun retention mechanism **106** comprises a first lever **104** pivotally mounted on the proximal wall portion **113** of the holster body **102** with a first axis of rotation. In embodiments, the first lever **104** has an actuation tab **107** rearwardly positioned on the lever **104** and positioned within the thumb button recess **218**. In embodiments, the actuation tab **107** has an upper margin **222**, a rearward margin **224**, and a lower margin **226**. In embodiments, the lever **104** further has a trigger guard blocking portion **128** forwardly positioned on the lever **104**, the lever **104** being configured and positioned such that depression of the actuation tab **107** rotates the lever **104** about its respective axis of rotation to move the trigger guard blocking portion **128** from a trigger guard blocking position to a non-blocking position.

(118) In embodiments, an upper portion **242** of the thumb button recess edge portion **220** extends rearwardly of the actuation tab **107** along the upper margin **222** of the actuation tab **107** and a lower portion **246** of the thumb button recess edge portion **220** extends rearwardly of the actuation tab **107** at the lower margin **226** of the actuation tab **107**.

(119) In embodiments, the holster **100** has only a single user actuatable handgun retention mechanism **106** and the actuation tab is configured as an actuation tab **107**. In embodiments, the holster **100** comprises a pair of thumb guides **230** positioned above and below the actuation tab **107**, the thumb guides **230** projecting laterally outward from the holster body proximal wall portion **113** beyond an undepressed position of the actuation tab **107**. In embodiments, the thumb guides **230** each extend laterally outward from an outward surface of the proximal wall portion **113** a distance of at least 0.35 inches.

(120) In embodiments, the holster **100** has a second handgun retention mechanism **174** including a pivoting shroud **176** positioned rearwardly of the slide **98** of the handgun **80** holstered in the holster **100**. In embodiments, the second handgun retention mechanism **174** comprises a second lever **104'**, the second lever **104'** having an actuation tab configured as a thumb button **180**, the actuation tab of the first lever position behind the thumb button **180** whereby depression of the thumb button **180** also depresses the actuation tab of the first lever.

(121) In embodiments, the holster **100** further comprises a pair of thumb guides **230** positioned above and below the thumb button **180**, the thumb guides **230** projecting laterally outward from the holster body **102** proximal wall portion **113** beyond an undepressed position of the thumb button **180**.

(122) In embodiments, one of the thumb guides **230** is positioned above the thumb button **180** and comprises a housing for the second handgun retention mechanism **174**.

(123) In embodiments, a belt engaging member **160** is attached to the holster body **102** with a plurality of threaded fasteners **162** extending into the holster body **102**.

(124) In embodiments, one of said threaded fasteners **162** is positioned above the first lever **104**, one of the plurality of threaded fasteners **162** is positioned below the first lever **104**, and one of the plurality of threaded fasteners **162** is positioned forward of the first lever **104**, wherein the first lever **104** is captured between the holster body **102** and the belt engaging member **160**.

(125) In embodiments, a column portion **236** of the belt engaging member **160** extends rearwardly, wherein when the handgun **80** is holstered. In embodiments, the thumb button **180** is positioned between the handgun body **84** and the belt engaging member **160**, wherein the belt engaging

member **160** has a pair of ribs **232** having their elongate dimension extending forwardly and rearwardly and positioned to confront the thumb guides **230** whereby access to the thumb button **180** is restricted on four sides of the thumb button **180** and an access path **252** for the thumb is provided rearwardly of the thumb button **180**. Ribs **232** and bottom rib **254** define a cavity **256** or recess with three sides in the column portion **236** of belt engaging member **160**. By providing a cavity **256** in belt engaging member **160**, an access path **252** is provided, while allowing the holster **100** to be positioned closer to the body of the user than if cavity **256** were not present and the fourth wall of the cavity **256** was instead provided by a surface coplanar with the rest of the column portion **236** of belt engaging member **160**. The ribs **232** may provide structural strengthening of the lower plate portion **258** as well as providing access inhibiting structure to prevent access by others from front of the holster **100** or the back side of the holster **100**, as the holster **100** is worn. The recess or cavity **256** opening upwardly providing access to the holster wearer as well as guide structure to easily and non-visually guide the user's thumb to the proper location both to actuate a thumb release button **180** and for the master grip.

(126) In embodiments, the belt engaging member **160** comprises a jacket slot belt loop **228**. In embodiments, the jacket slot belt loop **228** has a column portion **236** including a lower plate portion **258** having a holster attachment region **260**, a mid level portion **302**, and an upper portion **264** with a pair of clamping belt loop portions **234**. The lower plate portion **258** having apertures **266** for receiving threaded fasteners **162** for attachment of the holster body **102**. The mid level portion **302** may have a distal side **268** that cooperates with the holster body **102** to capture components of a retention mechanism **106** between the holster body **102** and the mid level portion **302** and thereabove a pair of upright ribs **232** protruding outwardly from the distal side **268** of mid level portion **302** defining recess or cavity **256** therebetween that extends upwardly to the upper portion **264** and is open upwardly. In embodiments, the jacket slot belt loop **228** having a pair of clamping belt loop portions **234** extending from the column portion **236**, each of the clamping belt loop portions **234** being adjustable with respect to the column portion **236** by a respective threaded fastener **262**. The upper portion **264** having a central column **272**. A pair of belt loop portions **234** are displaced distally inward of the central column **272** and displaced laterally from the central column **272**. The belt loop portions **234** connect to the central column **272** at upper connecting portions **274**. In certain embodiments, the central column **272**, the upper connection portions **274** and the belt portions **234** all unitary with each other. Each belt loop portion **234** having a lower end **276** not unitarily joined to the central column **272** but having a closable spacing therefrom. The central column **272** may have protruding portions configured as bosses **280** to cooperate with the lower ends **276** of the belt loop portions **234**. Threaded fasteners **262** may be utilized to adjust the spacing between each belt loop portions **234** and the central column **272** effecting a clamping action onto a belt, such as a duty belt, whereby the jacket slot belt loop **228** is secured to a user's belt. Optional elastomeric bushings (not shown) may be utilized between the lower ends **276** of the belt loop portions **234** and bosses **280** of the central column **272** wherein the bushings comprise a bore in a central axis of bosses **280** for receiving the threaded fasteners **262**. A feature and advantage of such embodiments is that the level of clamping of the jacket slot belt loop **228** and holster **100** are readily adjustable. Moreover, the lateral offset of the belt loop portions **234** from the central column **272** provides stability for the mounting system and holster **100** as the length of the engagement of the jacket slot belt loop **228** with the holster **100** is extended.

(127) In certain embodiments, threaded fasteners **262** can be unscrewed from lower ends **276** of the belt loop portions **234** such that there is a gap **284** between lower ends **276** of the belt loop portions **234** and bosses **280**. This gap **284** allows the jacket slot belt loop **228** to receive a belt of the user without the user needing to weave the belt through belt slots **286** formed by the openings in between belt loop portions **234** and central column **272**. This allows user to receive a belt in the belt slots **286** without removing other accessories already attached to the belt. Alternatively, a user can weave a belt through belt slots **286** without fully unscrewing threaded fasteners **262**. Once a

belt is received in the belt slots **286**, the threaded fasteners **262** can be re-engaged with the lower ends **276** of clamping belt loop portions **234**. The belt can be more tightly secured by tightening threaded fasteners **262**. In certain embodiments, if a belt is of a narrower width than belt slots **286**, the action of tightening threaded fasteners **262** may force the belt up in the belt slots **286** so that the belt is biased against the upper margin **288** of upper portion **292** of belt slots **286** (FIGS. **35A** and **35D**)**23**. In this manner, belt slots **286** can accommodate a belt that has a width less than that of belt slots **286**. In certain embodiments, the distance between the lower ends **290** of the belt slots **286** near the bosses **280** may be greater than the distance between the upper portion **292** of the belt slots **286** near upper margin **288**.

(128) In some embodiments, the outside face **294** of the clamping belt loop portions **234** may have protruding features **296**, such as ribs or studs, in order to receive a belt more securely (FIG. On other embodiments, the back side **298** of column portion **236** of belt engaging member **160** may also have protruding features **300** such as ribs or studs, in order to receive belt more securely.

(129) In embodiments, the thumb guides **230** each extend laterally outward from an outward surface of the proximal wall portion **113** a distance of at least 0.35 inches. In embodiments, the thumb guides **230** each extend laterally outward from an outward surface of the proximal wall portion **113** a distance of at least 0.35 inches and the ribs **232** extend outwardly from a surface of the column portion **236** a distance of at least 0.15 inches.

(130) In embodiments, the first lever **104** is seated in a lever receiving region **238** on the proximal wall portion **113** of the holster body and the proximal wall portion **113** has three threaded bosses **240** dispersed around the lever receiving region **238** for receiving a belt engaging member **160**. In embodiments, the proximal wall portion **113** further has a plurality of lever region containment wall portions **244** extending from the outer surface of the proximal wall portion **113** and each of the plurality of lever containment wall portions **244** connect to at least one of the three threaded bosses **240**. In embodiments, the three threaded bosses **240** are positioned for receiving a belt engaging member **160** utilizing a plurality of threaded fasteners **162**.

(131) In embodiments, the first lever **104** is seated in a lever receiving region **238** on the proximal wall portion **113** and the proximal wall portion **113** has three threaded bosses **240** dispersed around the lever receiving region **238** for receiving a belt engaging member **160** and, when the belt engaging member **160** is attached, the first lever **104** is captured within the lever receiving region **238**. In embodiments, the first lever **104** may be removed from an engagement with the proximal wall portion **113** when the belt engaging member is not attached to the proximal wall portion **113**. In embodiments, the first lever **104** may be removed without tools when the belt engaging member is not attached. In embodiments, the first lever **104** may be removed by simply prying the first lever **104** outward when the belt engaging member **160** is not attached.

(132) In embodiments, the holster body **102** has an upper wall portion having a pair of inclined wall portions defining a joint, a lower wall portion with a slit **250** extending from a forward opening to proximate a trigger guard receiving portion of the holster body defining a pair of forward clamping wall portions **248**. In embodiments, a threaded fastener **142** is positioned so as to extend between the pair of forward clamping wall portions **248** for adjusting the forward clamping wall portions **248** about a forward portion of the handgun **80**.

(133) Referring to FIGS. **1-47**, in embodiments, a holster **100** is provided for receiving a handgun **80** having a handgun body **84**, a slide **98**, and a trigger guard **90**. In embodiments, the holster **100** comprises a holster body **102** having a plurality of holster wall portions defining a rearward opening cavity **108** extending along a handgun receiving and withdrawal axis **110H** for receiving and holding the handgun **80**. In embodiments, the plurality of holster wall portions comprise a user proximal wall portion **113** and a user distal wall portion **115**. In embodiments, the holster includes a first handgun retention mechanism **106'** and a second handgun retention mechanism **174**, disposed at the proximal wall portion **113**, each handgun retention mechanism selectively preventing the handgun **80**, when in the cavity **108**, from being withdrawn from the cavity **108**. In embodiments,

the first and second handgun retention mechanisms have a respective first lever **104'** having a first axis of rotation and a second lever **180** with an axis of rotation, each lever having a respective actuation tab portion on one end of the lever where depression of the actuation tab rotates the lever about its respective axis of rotation, one of the two actuation tabs configured as a thumb receiving button with an outer thumb receiving surface, the other of the two actuation tabs positioned behind the thumb receiving button such that depression of the thumb receiving button effects a depression of the other of the two actuation tab portions, whereby depression of the thumb receiving button actuates both the first handgun retention mechanism **106'** and the second handgun retention mechanism **174**. In embodiments, the first lever **104'** of the first handgun retention mechanism **106'** is pivotally supported by the holster body **102**. In embodiments, the first lever **104'** comprises an elongate central portion **124** and a blocking portion **128** at an end opposite the respective actuation tab, the blocking portion **128** movable in and out of a handgun blocking position. In embodiments, the second handgun retention mechanism **174** comprising a U-shaped member **176** that is pivotally supported by the holster body **102** so that the U-shaped member **176** selectively pivots between a handgun retaining position in which the U-shaped member extends across a portion of a rearward opening of the cavity **108** and a release position in which the U-shaped member **176** is displaced from the portion of the rearward opening so that a withdrawal path of the handgun **80** is unobstructed by U-shaped member **176**.

(134) The following United States patents are hereby incorporated by reference herein: U.S. Pat. Nos. 5,048,735, 5,100,036, 5,129,562, 5,275,317, 5,284,281, 5,372,288, 5,395,021, 5,419,474, 5,449,103, 5,509,591, 5,573,157, 5,810,221, 5,810,221, 5,918,784, 5,918,784, 6,112,962, 6,189,751, 6,230,946, 6,267,279, 6,276,581, 6,533,149, 6,547,111, 6,547,111, 6,634,527, 6,641,009, 6,641,009, 6,752,300, 6,752,300, 6,769,582, 6,799,392, 6,854,626, 7,200,965, 7,434,712, 7,461,765, 7,461,765, 7,530,456, 7,530,456, 7,556,181, 7,556,181, 7,841,497, 7,841,497, 7,922,050, 7,922,050, 8,141,758, 8,141,758, 8,177,108, 8,235,263, 8,474,670, 8,517,235, 8,602,276, 8,602,276, 8,631,981, 8,631,981, 8,646,665, 8,720,753, 8,720,753, 8,720,754, 8,720,755, 8,851,344, 8,985,412, 9,022,262, 9,022,262, 9,057,579, 9,057,580, 9,134,093, 9,134,093, 9,175,925, 9,175,925, 9,228,802, 9,267,760, 9,347,741, 9,383,165, 9,410,767, 9,500,426, 9,777,986, and 9,835,400. Components illustrated in such patents may be utilized with embodiments herein. Incorporation by reference is discussed, for example, in MPEP section 2163.07(B).

(135) The patents and other references mentioned above in all sections of this application are herein incorporated by reference in their entirety for all purposes.

(136) All of the features disclosed in this specification (including the references incorporated by reference, including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive.

(137) Each feature disclosed in this specification (including references incorporated by reference, any accompanying claims, abstract and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

(138) The invention is not restricted to the details of the foregoing embodiment(s). The invention extends to any novel one, or any novel combination, of the features disclosed in this specification (including any incorporated by reference references, any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed. The above references in all sections of this application are herein incorporated by references in their entirety for all purposes.

(139) Although specific examples have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that any arrangement calculated to achieve the same purpose

could be substituted for the specific examples shown. This application is intended to cover adaptations or variations of the present subject matter. Therefore, it is intended that the invention be defined by the attached claims and their legal equivalents, as well as the following illustrative aspects. The above described aspects embodiments of the invention are merely descriptive of its principles and are not to be considered limiting. Further modifications of the invention herein disclosed will occur to those skilled in the respective arts and all such modifications are deemed to be within the scope of the invention.

Claims

1. A holster mount comprising: a column portion comprising: a lower portion comprising a holster attachment region, the holster attachment region comprising a distal side configured to attach to a holster; and at least one of a mid-level portion disposed vertically above the lower portion, and an upper portion disposed vertically above the mid-level portion, wherein the mid-level portion and the upper portion do not comprise the holster attachment region; the at least one of the mid-level portion and the upper portion comprising an upwardly extending thumb receiving recess defined by a protruding portion configured to protrude distally from the column portion towards and be adjacent to the holster attached to the holster attachment region, wherein the protruding portion is configured such that access to the recess by a user's thumb is not blocked from above when the holster is attached to the holster attachment region; and wherein the holster mount is configured to receive a user's belt.
2. The holster mount of claim 1, wherein the recess is configured to accommodate a user's thumb while drawing a handgun from the holster attached to the holster attachment region.
3. The holster mount of claim 1, the holster attachment region further comprising at least one aperture for receiving at least one threaded fastener configured for attachment to a holster body of the holster attached to the holster attachment region.
4. The holster mount of claim 1, the recess configured to cooperate with a holster body of the holster attached to the holster attachment region to capture components of a retention mechanism of the holster body between the holster body and the recess.
5. The holster mount of claim 1, further comprising at least one belt loop portion extending from the column portion and configured to receive a user's belt and, wherein the at least one belt loop portion is displaced distally inward of the column portion and displaced laterally from the column portion.
6. The holster mount of claim 5, wherein the at least one belt loop portion extends from the upper portion of the column portion.
7. The holster mount of claim 6, the upper portion further comprising an upper connection portion connecting the at least one belt loop portion to the upper portion; the upper portion, the upper connection portion, and the at least one belt loop portion being unitary with each other.
8. The holster mount of claim 5, wherein each of the at least one belt loop portion is configured for clamping.
9. The holster mount of claim 8, wherein each of the at least one belt loop portion is adjustable with respect to the column portion by a respective threaded fastener, wherein the at least one belt loop portion further comprises a lower end not unitarily joined to the column portion, but having a closable space therefrom, the closable space closable by the threaded fastener.
10. The holster mount of claim 9, the column portion further comprising protruding cooperating portions configured to cooperate with the lower ends of the at least one belt loop portion, wherein the protruding cooperating portions comprise bosses.
11. The holster mount of claim 5, the at least one belt loop portion further comprising an outside face, the outside face comprising protruding belt receiving features selected from a group consisting of ribs and studs.

12. The holster mount of claim 1, the column portion further comprising a back side, the back side comprising protruding belt receiving features selected from a group consisting of ribs and studs.

13. The holster mount of claim 1, wherein the protruding portion extends into at least a portion of the upper portion.

14. The holster mount of claim 13, wherein the protruding portion comprises a lower rib and two side ribs configured to be adjacent to a holster attached to the holster attachment region, wherein the side ribs extend into at least a portion of the upper portion of the column portion disposed above the holster attachment region.

15. A holster system for receiving a handgun, comprising: a holster comprising: a holster body having a plurality of holster wall portions defining a handgun receiving cavity; a handgun retention mechanism comprising a thumb actuator at one of the holster wall portions for selectively preventing the handgun, when in the cavity, from being withdrawn from the cavity; a holster mount attached to the holster comprising: a column portion comprising: a holster attachment region comprising a distal side configured to attach to the holster; a recess defined by a protruding portion configured to be adjacent to and protrude distally from the column portion towards the holster attached to the holster attachment region wherein the recess is disposed opposite the thumb actuator and is configured to accommodate a user's thumb while actuating the thumb actuator and drawing the handgun from the holster; and wherein the holster mount is configured to receive a user's belt.

16. The holster system of claim 15, wherein the column portion further comprises: a lower plate portion; a mid-level portion; and an upper portion; wherein the lower plate portion comprises the holster attachment region.

17. The holster system of claim 16, further comprising at least one belt loop portion configured to receive a user's belt and extending from the upper portion of the column portion, and wherein each of the at least one belt loop portions is configured for clamping and is adjustable with respect to the column portion by a respective threaded fastener.

18. The holster mount of claim 16, wherein the thumb receiving recess is defined by a lower rib and two side ribs configured to be adjacent to a holster attached to the holster attachment region.

19. The holster mount of claim 18, wherein the side ribs extend into at least a portion of the upper portion of the column portion.

20. A method of withdrawing a handgun from a holster attached to a holster mount, the method comprising: downwardly extending a user's thumb into a thumb receiving recess defined by a protruding portion configured to be adjacent to and protrude distally from a column portion of the holster mount towards a holster attached to a distal side of a holster attachment region of the column portion, wherein the protruding portion is configured such that access to the recess by a user's thumb is not blocked from above when the holster is attached to the holster attachment region; accessing a thumb actuator disposed opposite the thumb receiving recess with the user's thumb extending into the thumb receiving recess, actuating the thumb actuator with the user's thumb to release a handgun retention mechanism attached to the holster for selectively releasing the handgun from the handgun retention mechanism; and withdrawing the handgun from the holster by the user, wherein the user's thumb slides off of the thumb actuator and onto the handgun.
